Amazon Sales Data Analysis

Python | Pandas | NumPy | Data Cleaning & Aggregation

Portfolio Project • 2025 • Dataset: 1,465 Products | 9 Categories

Dataset Impact Summary

The following KPIs are derived directly from the amazon.csv dataset (1,465 product records across 9 categories). These metrics demonstrate the real-world scale and analytical value of the project.

Metric	Value	Metric	Value
Total Products	1,465	Avg Discount	47.7%
Total Categories	9	Max Discount	94%
Avg Rating	4.10 / 5.0	Products Rated 4+	1,110 (75.8%)
Avg MRP (Original)	₹5,445	Avg Sale Price	₹3,125
Avg Customer Saving	₹2,320	Total Reviews	26.7 Million

Project Overview

This project performs end-to-end exploratory data analysis (EDA) on an Amazon product catalogue dataset using Python. The goal is to clean, filter, aggregate, and export sales data to derive actionable business insights — identifying top-performing categories, understanding pricing and discount strategies, and analysing customer ratings at scale.

Tech Stack

Language	**Python 3.x**
Libraries	Pandas, NumPy
Data Source	Amazon product dataset (amazon.csv — 1,465 records)
Output	Processed Excel files (.xlsx)
IDE	Spyder (Anaconda)

Project Structure

```
AmazonSalesProject/
├── AmazonSalesProject.py                          # Main analysis script
├── amazon.csv / sales_data.xlsx                   # Raw input dataset
├── average_sales_by_category_and_status.xlsx     # Output 1
└── total_sales_by_ship_and_fulfil.xlsx            # Output 2
```

Category Analysis (From Dataset)

The dataset spans 9 main categories. The top 3 categories by product count account for 98% of the catalogue:

Category	Product Count	Avg Discount
Electronics	526 products (35.9%)	50.8%
Computers & Accessories	453 products (30.9%)	54.0%
Home & Kitchen	448 products (30.6%)	40.1%
Office Products	31 products (2.1%)	—
Others (5 categories)	7 products (0.5%)	—

Pricing & Discount Insights (From Dataset)

Insight	Finding	Business Impact
Average Discount Depth	47.7% across all products	Customers save nearly half the MRP on average
Electronics Savings	₹4,161 average saved per product	Highest absolute savings category
Home & Kitchen Savings	₹1,831 average saved per product	Mid-tier savings, high volume
Computers & Accessories	₹841 average saved per product	Lower-priced items, higher discount %
Price Range Distribution	39.4% products priced below ₹500	Affordable product mix dominates catalogue
Premium Segment (>10K)	122 products (8.3% of catalogue)	High-value electronics and appliances

Customer Rating Insights (From Dataset)

Metric	Value	Interpretation
Overall Avg Rating	4.10 out of 5.0	Strong customer satisfaction signal
Products with 4+ Stars	1,110 of 1,465 (75.8%)	Majority of catalogue is well-rated
Rating Band: 4.0 – 4.5	900 products (61.4%)	Core quality cluster
Rating Band: 3.5 – 4.0	467 products (31.9%)	Satisfactory but below benchmark
Computers & Accessories	Avg rating: 4.15	Highest-rated category
Home & Kitchen	Avg rating: 4.04	Slightly below Electronics
Total Customer Reviews	26.7 Million	Massive engagement and trust signal

Key Features & Implementation

1. Data Loading & Exploration

The dataset is loaded from CSV/Excel into a Pandas DataFrame. Initial exploration establishes structure, data types, and summary statistics before any cleaning begins.

```
sales_data = pd.read_excel('sales_data.xlsx')
sales_data.info()        # Column types, non-null counts
sales_data.describe()    # Mean, std, min, max per numeric column
print(sales_data.dtypes)
```

2. Data Cleaning

Out of 1,465 records, only 2 rows had missing rating_count values. Targeted cleaning on the Amount/Price column ensures no revenue metrics are skewed by nulls.

```
print(sales_data.isnull().sum())    # Only 2 nulls in rating_count
sales_data_clean = sales_data.dropna(subset=['Amount'])
```

3. Filtering & Slicing

Boolean indexing isolates high-value orders and specific category-quantity combinations for targeted analysis.

```
# High-value orders
```

```
high_amount_data = sales_data_clean[sales_data_clean['Amount'] > 1000]
# Multi-condition filter
sales_tops_3 = sales_data_clean[
    (sales_data_clean['Category'] == 'Tops') & (sales_data_clean['Qty'] == 3)]
```

4. Data Aggregation

Multi-level groupby operations produce category-level summaries across total sales, average order value, fulfilment method, and courier status.

```
# Total sales by category (descending)
category_totals = sales_data.groupby('Category',
    as_index=False)['Amount'].sum().sort_values('Amount', ascending=False)

# Avg by category + fulfilment
fulfillment_averages = sales_data.groupby(['Category','Fulfilment'],
    as_index=False)['Amount'].mean()
```

5. Data Export

```
status_averages.to_excel('average_sales_by_category_and_status.xlsx', index=False)
total_sales_shipandfulfil.to_excel('total_sales_by_ship_and_fulfil.xlsx',
index=False)
```

Skills Demonstrated

- **Data loading from CSV/Excel using Pandas read_csv() / read_excel()**
- **Exploratory Data Analysis (EDA): info(), describe(), dtypes, head()**
- **Data cleaning with isnull().sum() and targeted dropna(subset=[...])**
- **Boolean indexing: single and multi-condition filtering with & operator**
- **groupby() aggregation: sum(), mean() across single and multi-level groups**
- **sort_values() for descending rank ordering of aggregated results**
- **Column renaming with rename(columns={...}, inplace=True)**
- **Exporting cleaned DataFrames to Excel with to_excel(index=False)**
- **Price cleaning: stripping currency symbols (₹) and commas with regex**
- **Feature engineering: extracting main/sub categories from pipe-delimited strings**

How to Run

Prerequisites:

```
pip install pandas numpy openpyxl
```

Steps:

- Clone this repository from GitHub

- Place amazon.csv or sales_data.xlsx in the project root directory
- Run: python AmazonSalesProject.py
- Output Excel files are generated in the same directory

Future Improvements

- Add data visualisations using Matplotlib or Seaborn (bar charts, heatmaps, scatter)
- Build an interactive dashboard using Plotly Dash or Power BI
- Extend analysis to sub-category level (Accessories, Kitchen Appliances, etc.)
- Integrate with SQL database for scalable data storage and querying
- Apply machine learning to predict discount ranges based on product attributes
- Sentiment analysis on review_content column using NLP techniques

Author

This project was developed as a Python data analysis portfolio to demonstrate practical data wrangling and business intelligence skills using real-world Amazon e-commerce data.

GitHub Repository: https://github.com/rajarshisadhya/python-fordataanalysis